ESAT PRACTICE TEST SERIES

CHEMISTRY + BIOLOGY

LEVEL 1 · STANDARD

Chemistry + Biology

Mathematics 1 + Chemistry + Biology

Representative ESAT-style breadth with routine and multi-step applications.

MODULE

Mathematics 1

MODULE

Chemistry

MODULE

Biology

FULL SOLUTION BOOK

ThrivingScholars

81 preserved questions · three verified module keys

COMBINED ANSWER KEYS

Chemistry + Biology · Level 1 Standard

This book compiles three existing 27-question modules into one 81-question pathway. No new questions have been generated and no source solution has been rewritten.

Composition: Mathematics 1 + Chemistry + Biology. Each module retains its original question order, answer and worked solution.

Mathematics 1

Engineering Mock Test 3 · preserved Mathematics 1 module

Q1 C	Q2 B	Q3 E	Q4 C	Q5 D
Q6 D	Q7 E	Q8 A	Q9 A	Q10 D
Q11 D	Q12 C	Q13 B	Q14 A	Q15 B
Q16 E	Q17 E	Q18 E	Q19 D	Q20 E
Q21 D	Q22 D	Q23 C	Q24 D	Q25 D
Q26 A	Q27 C			

Chemistry

Preserved Chemistry Standard source set

Q1 C	Q2 D	Q3 B	Q4 C	Q5 E
Q6 B	Q7 E	Q8 C	Q9 D	Q10 D
Q11 E	Q12 F	Q13 B	Q14 E	Q15 C
Q16 B	Q17 D	Q18 C	Q19 A	Q20 C
Q21 B	Q22 C	Q23 D	Q24 C	Q25 A
Q26 E	Q27 C			

Biology

Preserved Biology Standard source set

Q1 D	Q2 C	Q3 C	Q4 G	Q5 C
Q6 F	Q7 E	Q8 F	Q9 C	Q10 D
Q11 B	Q12 E	Q13 B	Q14 D	Q15 C
Q16 E	Q17 E	Q18 H	Q19 B	Q20 C
Q21 E	Q22 D	Q23 B	Q24 B	Q25 C
Q26 F	Q27 E			

LEVEL 1 · STANDARD

Mathematics 1

27 QUESTIONS · COMPLETE SOLUTIONS

Engineering Mock Test 3 · preserved Mathematics 1 module. Question wording, option order, answer and solution method are unchanged.

M1-01

Mock Test 3 - Mathematics 1

Answer C

QUESTION 1

There are 8 consecutive integers. If the sum of the first three is 63, what is the largest one?

- **A** 17
- **B** 25
- **C** 27
- **D** 30
- **E** 31

WORKED SOLUTION

Let the first integer be n. Then $n + (n + 1) + (n + 2) = 63$, so $3n + 3 = 63$, giving $n = 20$. The largest of the eight integers is 27.

M1-02

Mock Test 3 - Mathematics 1

Answer **B**

QUESTION 2

Joshua is 10 years old and his father is 42 years old. After how many years, his father's age is three times Joshua's age?

- **A** 5
- **B** 6
- **C** 7
- **D** 8
- **E** 9

WORKED SOLUTION

After t years, $42 + t = 3(10 + t)$. Hence $42 + t = 30 + 3t$, so $t = 6$.

M1-03

Mock Test 3 - Mathematics 1

Answer **E**

QUESTION 3

N is a 7-digit number consists of 2 or 3. There are more 2s than 3s in N. If N is divisible by both 3 and 4, what is the remainder of N when it is divided by 7?

- **A** 1
- **B** 2
- **C** 3

D 4

E 5

WORKED SOLUTION

Let k be the number of 3s. The digit sum is $14 + k$, so divisibility by 3 gives $k = 1$. Divisibility by 4 forces the last two digits to be 32. Thus $N = 2222232$, and $N \equiv 5 \pmod{7}$.

M1-04

Mock Test 3 - Mathematics 1

Answer C

QUESTION 4

The preliminary round of a violin competition is divided into four rounds. According to the rule, the average score of a percipient of these four rounds must be not less than 96 points in order to enter the finals. Shaelyn's scores in the first three rounds are 95, 97, and 94. What is the minimum score that she needs to get on the fourth round in order to enter the finals? (Hope Cup Math Competition, China)

A 96

B 97

C 98

D 99

E 100

WORKED SOLUTION

The total score needed is $4 \times 96 = 384$. The first three scores total $95 + 97 + 94 = 286$, so the fourth score must be at least $384 - 286 = 98$.

M1-05 Mock Test 3 - Mathematics 1 **Answer D**

QUESTION 5

A haunted house has 8 windows. In how many ways can George the Ghost enter the house by one window and leave by a different window?

- **A** 28
- **B** 32
- **C** 40
- **D** 56
- **E** 63

WORKED SOLUTION

There are 8 choices for the entry window and then 7 different choices for the exit window, giving $8 \times 7 = 56$.

M1-06 Mock Test 3 - Mathematics 1 **Answer D**

QUESTION 6

Jackie bought a bottle (30 oz) of 90% alcohol from pharmacy to make homemade disinfectant spray. How many ounces of water should he add in order to dilute it into 75% alcohol? (Hint: The amount of alcohol remains the same.)

A 3

B 4

C 5

D 6

E 7

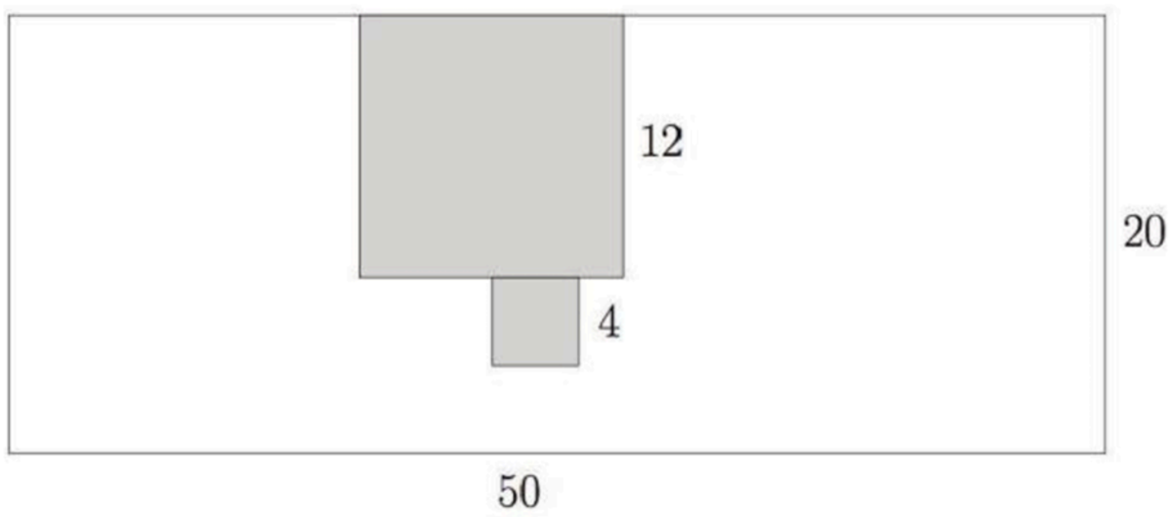
WORKED SOLUTION

The amount of alcohol is $30 \times 0.90 = 27$ oz. If the final concentration is 75%, then $0.75\mathrm{T} = 27$, so $\mathrm{T} = 36$. Water added: $36 - 30 = 6$ oz.

M1-07 Mock Test 3 - Mathematics 1 Answer E

QUESTION 7

Cut a 12×12 square and a 4×4 square from a 50×20 rectangle paper. What is the perimeter of the remaining part?



A 128

B 140

C 152

D 160

E 172

WORKED SOLUTION

The original rectangle perimeter is $2(50+20)=140$. The 12×12 notch increases the perimeter by 24, and the attached 4×4 cut-out adds another 8. Total perimeter $=140+32=172$.

M1-08 Mock Test 3 - Mathematics 1 Answer A

QUESTION 8

Jasmine keeps adding numbers one by one from the following list

1, 3, 5, 7, $\cdots$.

as $1+3=4$, $1+3+5=9$, $1+3+5+7=16$, etc. At one point, she got 379 and realized that one number is missed in the addition. What is the sum of digits of the missing number?

A 3

B 6

C 8

D 9

E 11

WORKED SOLUTION

The sum of the first n odd numbers is n^2. Since 379 is just below $20^2 = 400$, the missing number is $400 - 379 = 21$. The digit sum is $2 + 1 = 3$.

M1-09

Mock Test 3 - Mathematics 1

Answer A

QUESTION 9

$\overline{ab}$ and $\overline{cd}$ are two 2-digit numbers. If

$$\overline{ab} - \overline{cd} = 24$$

and

$$\overline{1ab} - \overline{cd1} = 15,$$

what is $a + d$?

A 5

B 8

C 10

D 11

E 12

WORKED SOLUTION

Using place value, $\overline{ab} = 10a + b$ and $\overline{cd} = 10c + d$. Solving the two given equations gives $a + d = 5$.

M1-10

Mock Test 3 - Mathematics 1

Answer **D**

QUESTION 10

The speed of a train is 15 meters per second. If it takes the train 10 seconds to pass a lamp post, how long does it take the train to completely pass a 300-meter long tunnel, in seconds? (Hint: first figure out the length of the train based on the information related to the lamppost)

A 10

B 20

C 25

D 30

E 35

WORKED SOLUTION

The train length is $15 \times 10 = 150$ m. To pass the tunnel completely, it travels $300 + 150 = 450$ m. Time $= 450/15 = 30$ s.

M1-11

Mock Test 3 - Mathematics 1

Answer **D**

QUESTION 11

How many pairs of positive integers (x, y) satisfy that $\frac{x}{3} = \frac{4}{y}$?

A 3

B 4

C 5

D 6

E 7

WORKED SOLUTION

From $x/3 = 4/y$, we get $xy = 12$. The positive factor pairs are $(1, 12), (2, 6), (3, 4), (4, 3), (6, 2), (12, 1)$, so there are 6 pairs.

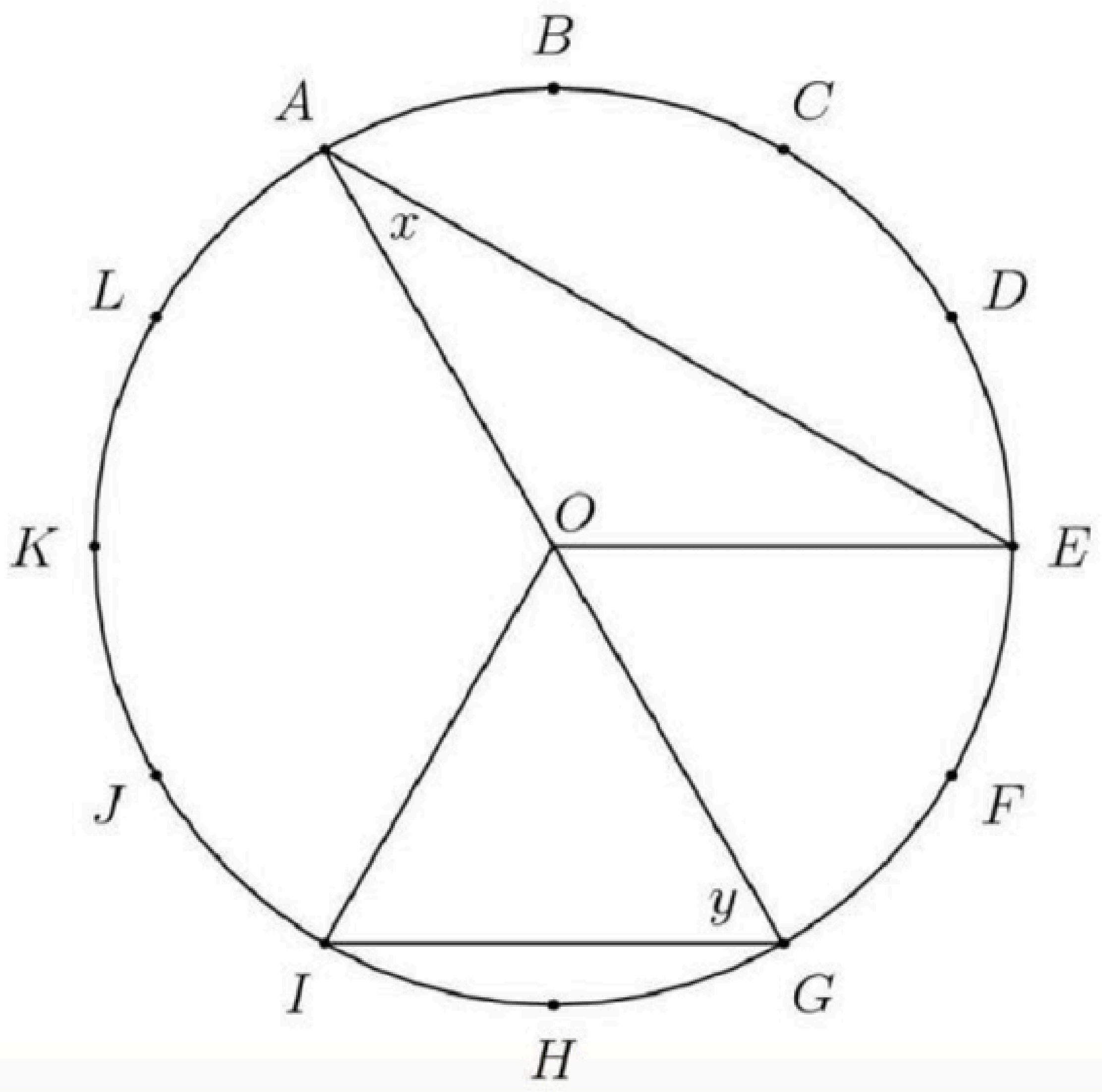
M1-12

Mock Test 3 - Mathematics 1

Answer C

QUESTION 12

The circumference of the circle with center O is divided into 12 equal arcs, marked the letters A through L as seen below. What is the number of degrees in the sum of the angles x and y?



A 75

B 80

C 90

D 120

E 150

WORKED SOLUTION

The circle is divided into 12 equal arcs, so each arc is 30°. Using central and inscribed angle relationships from the diagram gives $x + y = 90^\circ$.

M1-13

Mock Test 3 - Mathematics 1

Answer B

QUESTION 13

If for whole numbers x, y and z,

$360 = 2^x \times 3^y \times 5^z$,

what is $x + y + z$? Here $a^n = a \times a \times \cdots \times a$.

For example, $100 = 2 \times 2 \times 5 \times 5 = 2^2 \times 5^2$.

A 5

B 6

C 7

D 8

E 9

WORKED SOLUTION

Prime factorise: $360 = 36 \times 10 = 2^2 \times 3^2 \times 2 \times 5 = 2^3 \times 3^2 \times 5^1$. Hence $x + y + z = 3 + 2 + 1 = 6$.

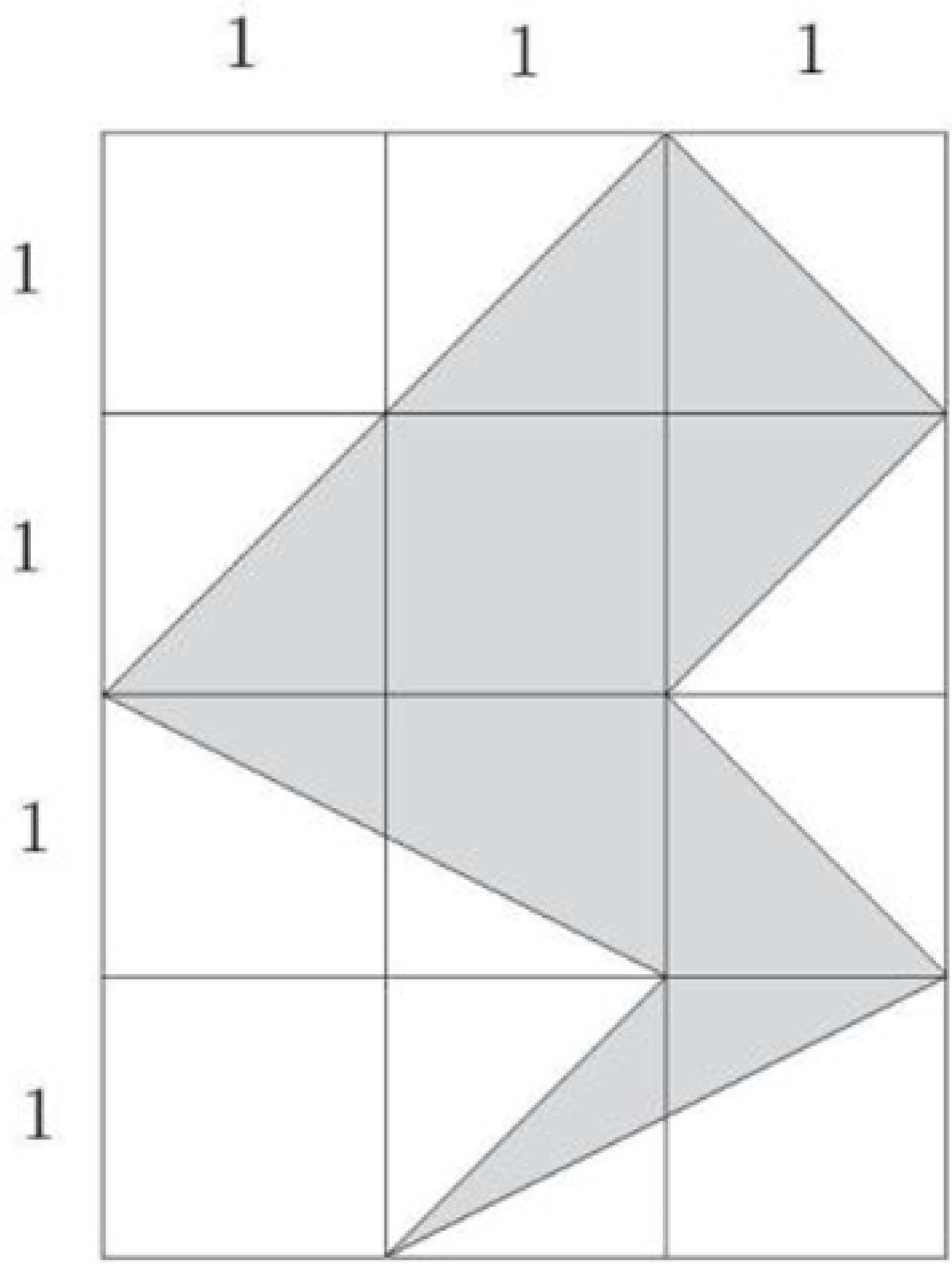
M1-14

Mock Test 3 - Mathematics 1

Answer A

QUESTION 14

What is the area of the shaded region in the following 3×4 grid?



A 5

B 6

C 7

D 8

E 9

WORKED SOLUTION

Counting the shaded polygon by subtracting the surrounding right triangles from the 3×4 rectangle gives shaded area 5.

M1-15 Mock Test 3 - Mathematics 1 Answer B

QUESTION 15

Randomly choose 3 different numbers from the set $\{1, 3, 5, 7, 9\}$. What is the probability that the three selected numbers could be side lengths of a triangle?

A $\frac{1}{5}$

B $\frac{3}{10}$

C $\frac{1}{3}$

D $\frac{2}{5}$

E $\frac{1}{2}$

WORKED SOLUTION

There are $\binom{5}{3} = 10$ triples. The valid triangle triples are $(3, 5, 7), (3, 7, 9), (5, 7, 9)$, so the probability is $3/10$.

M1-16

Mock Test 3 - Mathematics 1

Answer E

QUESTION 16

Marley practices exactly one sport each day of the week. She runs three days a week but never on two consecutive days. On Monday she plays basketball and two days later golf. She swims and plays tennis, but she never plays tennis the day after running or swimming. Which day of the week does Marley swim?

- **A** Sunday
- **B** Tuesday
- **C** Thursday
- **D** Friday
- **E** Saturday

WORKED SOLUTION

Monday is basketball and Wednesday is golf. Applying the non-consecutive running rule and the condition that tennis is not immediately after running or swimming leaves Saturday as the swimming day.

M1-17

Mock Test 3 - Mathematics 1

Answer E

QUESTION 17

Three toy cars run around a round track. It takes them 8, 10 and 12 seconds to finish one full circle respectively. If three cars start together from the same point and run toward the same direction, after how many seconds will they get back to the starting point together for the first time?

A 20

B 40

C 60

D 80

E 120

WORKED SOLUTION

The cars return together after the least common multiple of $8, 10, 12$, which is 120 seconds.

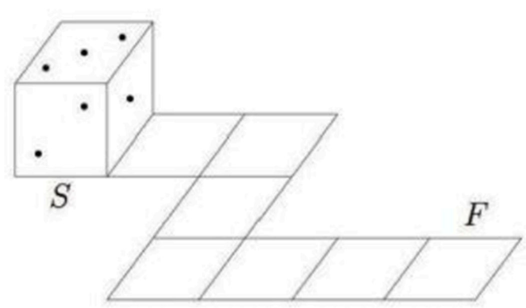
M1-18

Mock Test 3 - Mathematics 1

Answer E

QUESTION 18

Opposite faces of a die always add to 7. A die rolls on a circuit as represented below. At the starting point (S) the top face is 3. Which will be the top face at the final point (F)?



A 2

B 3

C 4

D 5

E 6

F 7

G 8

H 9

WORKED SOLUTION

Track the die through the path, using opposite faces summing to 7 after each roll. The final top face is 6.

M1-19

Mock Test 3 - Mathematics 1

Answer D

QUESTION 19

How many numbers x satisfy that $(x^2 - 2)^2 = 4$?

A 0

B 1

C 2

D 3

E 4

F 5

G 6

H 7

WORKED SOLUTION

$(x^2 - 2)^2 = 4$ gives $x^2 - 2 = \pm 2$. Thus $x^2 = 4$ or $x^2 = 0$, giving $x = -2, 0, 2$. There are 3 real numbers.

M1-20

Mock Test 3 - Mathematics 1

Answer E

QUESTION 20

For how many two-digit numbers, if you switch the tens and ones digits (e.g., 18 → 81), the new 2-digit number is 18 more than the original one?

A 3

B 4

C 5

D 6

E 7

WORKED SOLUTION

Let the number be $10a + b$. Reversing digits gives $10b + a$. The condition gives $10b + a = 10a + b + 18$, so $b - a = 2$. There are 7 possible tens digits.

M1-21

Mock Test 3 - Mathematics 1 | Answer **D**

QUESTION 21

[AMC 8 Problem] Malcolm wants to visit Isabella after school today and knows the street where she lives but doesn't know her house number. She tells him, "My house number has two digits, and exactly three of the following four statements about it are true."

- It is prime.
- It is even.
- It is divisible by 7.
- One of its digits is 9.

This information allows Malcolm to determine Isabella's house number. What is its units digit?

A 4

B 6

C 7

D 8

E 9

WORKED SOLUTION

The number must have exactly three of the four properties. The number 98 is even, divisible by 7, has a digit 9, and is not prime. Its units digit is 8.

M1-22

Mock Test 3 - Mathematics 1

Answer **D**

QUESTION 22

In the addition problem shown, m, n, p, and q represent positive digits. When the problem is completed correctly, the value of $m + n + p + q$ is (Canadian Math Competition) t is prime.

t is even.

t is divisible by 7.

)ne of its digits is 9.

information allows M

- **A** 16
- **B** 18
- **C** 22
- **D** 24
- **E** 30

WORKED SOLUTION

From the units column, $3 + 2 + q$ ends in 2, so $q = 7$ with carry 1. From the tens column, $6 + p + 8 + 1$ ends in 4, so $p = 9$. From the hundreds column,

$n + 7 + 5 + 2$ ends in 0, so $n = 6$ with carry $m = 2$. Therefore $m + n + p + q = 2 + 6 + 9 + 7 = 24$.

M1-23

Mock Test 3 - Mathematics 1

Answer **C**

QUESTION 23

If two numbers x and y satisfy that $(x + 1)^2 + (y - 5)^2 = 0$, what is $x + y$?

- **A** −1
- **B** 0
- **C** 4
- **D** 6
- **E** 7

WORKED SOLUTION

A sum of squares is zero only if both squares are zero. Thus $x + 1 = 0$ and $y - 5 = 0$, so $x = -1, y = 5$, and $x + y = 4$.

M1-24

Mock Test 3 - Mathematics 1

Answer **D**

QUESTION 24

Abel tosses two coins and Ben tosses one coin. What is the probability that they get the equal number of heads?

Recall that

$$\text{Probability} = \frac{\text{Number of valid cases}}{\text{Number all possible cases}}.$$

A $\frac{1}{8}$

B $\frac{1}{4}$

C $\frac{1}{3}$

D $\frac{3}{8}$

E $\frac{2}{5}$

WORKED SOLUTION

There are 8 equally likely outcomes. Equal heads occurs when Abel gets 0 heads and Ben gets 0 heads, or when Abel gets 1 head and Ben gets 1 head: $1 + 2 = 3$ cases. Probability $= 3/8$.

M1-25

Mock Test 3 - Mathematics 1

Answer D

QUESTION 25

x, y, z and w are four numbers. If the average of the following three numbers $x - y,\ y - z,\ z - w$ is 10 and $w = 5$. What is x?

A 15

B 20

C 25

D 35

E there is no enough information

WORKED SOLUTION

The average of $x - y, y - z, z - w$ is 10, so their sum is 30. The sum telescopes: $(x - y) + (y - z) + (z - w) = x - w = 30$. Since $w = 5$, $x = 35$.

M1-26 Mock Test 3 - Mathematics 1 Answer A

QUESTION 26

What is the tens digit of 7^{2020}?

A 0

B 1

C 2

D 3

E 4

WORKED SOLUTION

The last two digits of powers of 7 cycle: $07, 49, 43, 01$. Since 2020 is divisible by 4, 7^{2020} ends in 01, so the tens digit is 0.

M1-27

Mock Test 3 - Mathematics 1

Answer C

QUESTION 27

Two trains 300 miles apart are traveling toward each other along the same track. One train goes 70 miles per hour; the other runs at 80 miles per hour. If both of them leave their stations at 7:00am, when will they meet?

A 8 : 00 am

B 8 : 30 am

C 9 : 00 am

D 9 : 30 am

E 10 : 00 am

WORKED SOLUTION

The trains approach each other at $70 + 80 = 150$ miles per hour. Time $= 300/150 = 2$ hours, so they meet at 9:00 am.

LEVEL 1 · STANDARD

Chemistry

27 QUESTIONS · COMPLETE SOLUTIONS

Preserved Chemistry Standard source set. Question wording, option order, answer and solution method are unchanged.

C-01

Atomic structure

Answer C

QUESTION 1

Which species has the **same number of neutrons and the same number of electrons** as ${}^{40}_{20}\mathrm{Ca}^{2+}$?

A ${}^{35}_{17}\mathrm{Cl}^{-}$

B ${}^{41}_{19}\mathrm{K}^{+}$

C ${}^{38}_{18}\mathrm{Ar}$

D ${}^{40}_{20}\mathrm{Ca}$

E ${}^{36}_{18}\mathrm{Ar}$

WORKED SOLUTION

For ${}^{40}_{20}\mathrm{Ca}^{2+}$, the number of neutrons is

$$40 - 20 = 20,$$

and the number of electrons is

$$20 - 2 = 18.$$

For ${}^{38}_{18}\mathrm{Ar}$,

$$38 - 18 = 20 \text{ neutrons}, \qquad 18 \text{ electrons}.$$

Both values match, so the correct answer is **C**.

C-02

Isotopes and relative atomic mass

Answer **D**

QUESTION 2

Element X occurs naturally as three isotopes of mass numbers 24, 25 and 26.

Their relative abundances are $6 : 1 : 1$.

What is the relative atomic mass of X?

- **A** 24.250
- **B** 24.500
- **C** 24.625
- **D** 24.375
- **E** 25.000

WORKED SOLUTION

Use the weighted mean:

$$\mathrm{A_r} = \frac{6(24) + 1(25) + 1(26)}{6 + 1 + 1} = \frac{195}{8} = 24.375.$$

Therefore the correct answer is **D**.

C-03

Periodicity and reactivity

Answer B

QUESTION 3

Four atoms have the electron configurations shown.

atom	electron configuration
W	2,7
X	2,8,1
Y	2,8,7
Z	2,8,8,1

Which pair would be expected to react **most vigorously with each other** to form an ionic compound?

A W and X

B W and Z

C X and Y

D Y and Z

E W and Y

WORKED SOLUTION

W and Y are Group 17 atoms; X and Z are Group 1 atoms.

Group 1 reactivity increases down the group, so Z is more reactive than X. Group 17 reactivity decreases down the group, so W is more reactive than Y.

The most vigorous pairing is therefore the most reactive metal with the most reactive halogen: **W and Z**.

C-04

Structure and bonding

Answer C

QUESTION 4

The properties of five substances are shown.

substance	melting point / °C	conducts when solid	conducts when molten
A	−20	no	no
B	660	good	good
C	805	no	good
D	1710	no	no
E	−114	no	no

Which substance is most likely to have a **giant ionic structure**?

- **A** A
- **B** B
- **C** C
- **D** D
- **E** E

WORKED SOLUTION

A giant ionic solid normally has a high melting point. It does not conduct when solid because the ions cannot move, but it conducts when molten because the ions become mobile.

Only substance C has this combination of properties, so the answer is **C**.

C-05

Reaction stoichiometry

Answer E

QUESTION 5

The reaction

$$2A(aq) + B(aq) \rightarrow C(aq)$$

goes to completion.

Initially, $[A] = 0.80 \text{ mol dm}^{-3}$ and $[B] = 0.50 \text{ mol dm}^{-3}$, and no C is present. The volume does not change during the reaction.

What are the final concentrations of B and C?

- **A** $[B] = 0.00,\ [C] = 0.50 \text{ mol dm}^{-3}$
- **B** $[B] = 0.10,\ [C] = 0.20 \text{ mol dm}^{-3}$
- **C** $[B] = 0.20,\ [C] = 0.40 \text{ mol dm}^{-3}$
- **D** $[B] = 0.10,\ [C] = 0.80 \text{ mol dm}^{-3}$
- **E** $[B] = 0.10,\ [C] = 0.40 \text{ mol dm}^{-3}$

WORKED SOLUTION

The reacting ratio is $2A : B : C = 2 : 1 : 1$.

Using all 0.80 mol dm^{-3} of A consumes

$$0.80 \div 2 = 0.40 \text{ mol dm}^{-3}$$

of B and forms the same concentration, 0.40 mol dm^{-3}, of C.

So

$$[B]_{\text{final}} = 0.50 - 0.40 = 0.10, \qquad [C]_{\text{final}} = 0.40.$$

Therefore the answer is **E**.

C-06

Equilibrium

Answer B

QUESTION 6

For a reversible gaseous reaction:

- increasing the temperature **decreases** the equilibrium yield of R;
- increasing the pressure **increases** the equilibrium yield of R.

Which equation could represent the reaction?

A $P + Q \rightleftharpoons 2R \qquad \Delta H < 0$

B $2P + Q \rightleftharpoons R + S \qquad \Delta H < 0$

C $2P \rightleftharpoons 3R \qquad \Delta H < 0$

D $2P + Q \rightleftharpoons R + S \qquad \Delta H > 0$

E $P + 2Q \rightleftharpoons 3R \qquad \Delta H < 0$

F $3P \rightleftharpoons R + 2S \qquad \Delta H > 0$

WORKED SOLUTION

If increasing temperature decreases the product yield, the forward reaction must be exothermic, so $\Delta H < 0$.

If increasing pressure increases the product yield, the product side must contain fewer moles of gas.

In B there are 3 gaseous particles on the left and 2 on the right, and the forward reaction is exothermic. Therefore **B** is possible.

C-07

Redox and oxidation states

Answer E

QUESTION 7

Chlorine reacts with cold aqueous hydroxide ions:

$$Cl_2 + 2OH^- \rightarrow Cl^- + ClO^- + H_2O$$

Consider the statements.

1. The oxidation state of chlorine in Cl_2 is 0.
2. The oxidation state of chlorine in ClO^- is +1.
3. Cl_2 acts only as an oxidising agent in this reaction.

Which statements are correct?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Elemental chlorine has oxidation state 0, so statement 1 is true.

In ClO^-, if chlorine has oxidation state x,

$$x - 2 = -1 \Rightarrow x = +1,$$

so statement 2 is true.

Some chlorine is reduced from 0 to −1 and some is oxidised from 0 to +1. Chlorine therefore acts as both an oxidising and a reducing agent, so statement 3 is false.

The answer is **E: 1 and 2 only**.

C-08

Chromatography

Answer **C**

QUESTION 8

Four dyes are separated using the same solvent. The solvent front moves $8.0\ \mathrm{cm}$.

dye	distance moved / cm
W	2.0
X	3.6
Y	5.2
Z	6.4

In a second experiment, an unknown dye moves $3.25\ \mathrm{cm}$ while the solvent front moves $5.0\ \mathrm{cm}$.

Which dye is the unknown most likely to be?

A W

B X

C Y

D Z

E It cannot be determined.

WORKED SOLUTION

For the unknown dye,

$$R_f = \frac{3.25}{5.0} = 0.65.$$

For dye Y,

$$R_f = \frac{5.2}{8.0} = 0.65.$$

So the unknown most likely corresponds to **Y**, answer **C**.

C-09

Empirical and molecular formulae

Answer **D**

QUESTION 9

A compound contains by mass:

- 40.0% carbon,
- 6.7% hydrogen,
- 53.3% oxygen.

Its relative molecular mass is 60.

$$A_r: \quad C = 12,\ H = 1,\ O = 16$$

What is its molecular formula?

A CH_2O

B $C_2H_2O_2$

C C_2H_4O

D $C_2H_4O_2$

E $C_3H_6O_3$

WORKED SOLUTION

For a 100 g sample, the amounts are approximately

$$C : \frac{40.0}{12} = 3.33, \qquad H : \frac{6.7}{1} = 6.7, \qquad O : \frac{53.3}{16} = 3.33.$$

Dividing by the smallest gives approximately $1 : 2 : 1$, so the empirical formula is CH_2O.

Its empirical formula mass is 30. Since $60/30 = 2$, the molecular formula is

$$\boxed{C_2H_4O_2}.$$

Therefore the answer is **D**.

C-10

Limiting reagent and percentage yield

Answer **D**

QUESTION 10

Aluminium reacts with oxygen.

$$4Al + 3O_2 \rightarrow 2Al_2O_3$$

2.70 g of aluminium is mixed with 4.80 g of oxygen. The mass of aluminium oxide actually produced is 4.08 g.

$$A_r: \quad Al = 27, \; O = 16$$

What is the percentage yield?

A 40%

B 60%

C 75%

D 80%

E 85%

F 90%

WORKED SOLUTION

$$n(Al) = \frac{2.70}{27} = 0.100 \text{ mol}, \qquad n(O_2) = \frac{4.80}{32} = 0.150 \text{ mol}.$$

Only 0.075 mol of O_2 is required for 0.100 mol of Al, so aluminium is limiting.

From the equation, 0.100 mol of Al can form 0.050 mol of Al_2O_3.

$$M_r(Al_2O_3) = 102, \qquad m_{\text{theoretical}} = 0.050(102) = 5.10 \text{ g}.$$

Thus

$$\% \text{ yield} = \frac{4.08}{5.10} \times 100 = 80\%.$$

The answer is **D**.

C-11

Strong and weak acids

Answer E

QUESTION 11

Solutions P and Q are both $0.10\ \mathrm{mol\ dm^{-3}}$ solutions of monoprotic acids.

- P is a strong acid.
- Q is a weak acid.

Consider the statements.

1. P has a lower pH than Q.
2. Equal volumes of P and Q require the same amount of sodium hydroxide for complete neutralisation.
3. Q has the greater electrical conductivity.

Which are correct?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

The strong acid is more completely ionised, so it has the larger $[H^+]$ and hence the lower pH. Statement 1 is true.

Both acids are monoprotic and have the same formal concentration. Equal volumes therefore contain the same amount of acid and require the same amount of OH^- for complete neutralisation. Statement 2 is true.

The strong acid contains more mobile ions and therefore has the greater conductivity, so statement 3 is false.

The answer is **E: 1 and 2 only**.

C-12

Titration | Answer F

QUESTION 12

A $1.56\ \mathrm{g}$ sample of a pure monoprotic acid HX is dissolved in water.

It requires exactly $25.0\ \mathrm{cm^3}$ of $0.80\ \mathrm{mol\ dm^{-3}}$ sodium hydroxide for complete neutralisation.

$$HX + NaOH \rightarrow NaX + H_2O$$

What is the relative molecular mass of HX?

- **A** 39
- **B** 52
- **C** 65
- **D** 72
- **E** 104
- **F** 78

WORKED SOLUTION

The amount of sodium hydroxide used is

$$n = CV = 0.80 \times 0.0250 = 0.0200 \text{ mol.}$$

The reaction ratio is 1:1, so the sample contains 0.0200 mol of HX.

Therefore

$$M_r(HX) = \frac{1.56}{0.0200} = 78.$$

The answer is **F**.

C-13

Rates of reaction

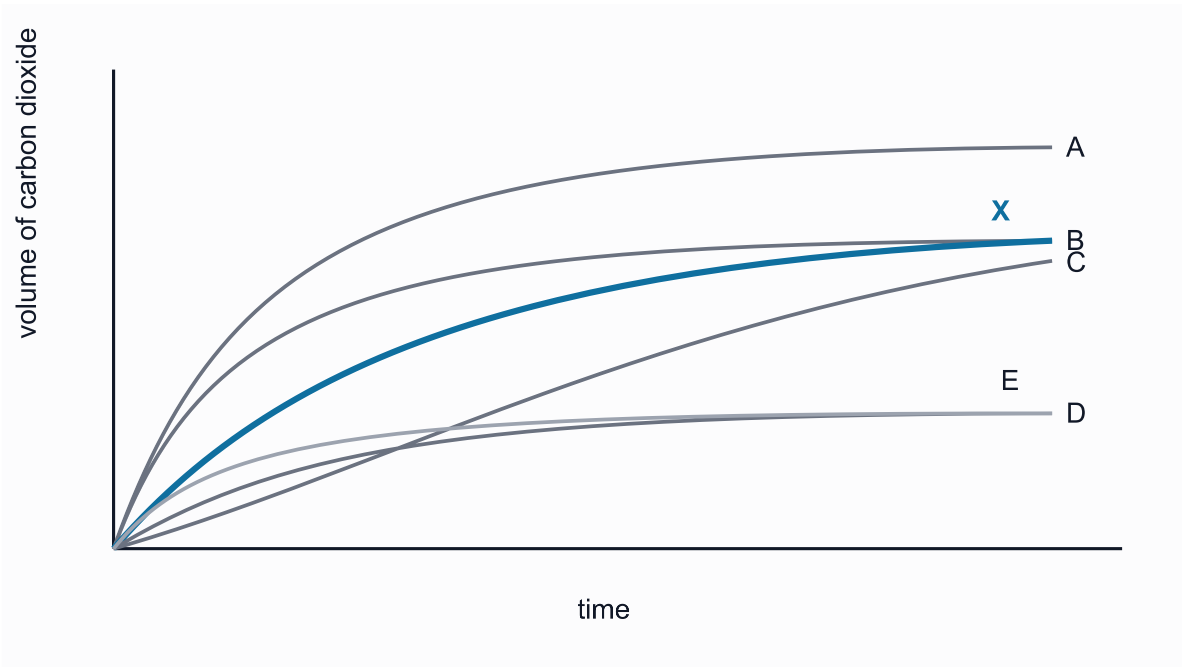
Answer **B**

QUESTION 13

Excess calcium carbonate reacts with 100 cm^3 of 1.0 mol dm^{-3} hydrochloric acid. Curve X shows the volume of carbon dioxide produced against time.

The experiment is repeated using 50 cm^3 of 2.0 mol dm^{-3} hydrochloric acid, with the same mass and particle size of excess calcium carbonate.

Which candidate curve represents the second experiment?



A curve A

B curve B

C curve C

D curve D

E curve E

WORKED SOLUTION

The amount of acid in the first experiment is

$$1.0 \times 0.100 = 0.100 \text{ mol.}$$

In the second experiment it is

$$2.0 \times 0.050 = 0.100 \text{ mol.}$$

So the same total amount of hydrochloric acid is available. Since calcium carbonate is in excess, the final amount of CO_2 is unchanged.

The second solution has twice the acid concentration, so the initial collision frequency and reaction rate are greater. The required curve must therefore reach the **same plateau more quickly**.

That is **curve B**.

C-14

Activation energy

Answer E

QUESTION 14

For a reaction,

$$\Delta H = -60 \text{ kJ mol}^{-1}.$$

The activation energy of the forward reaction is 90 kJ mol^{-1}.

What is the activation energy of the reverse reaction?

A $-150\ \text{kJ mol}^{-1}$

B $-30\ \text{kJ mol}^{-1}$

C $30\ \text{kJ mol}^{-1}$

D $90\ \text{kJ mol}^{-1}$

E $150\ \text{kJ mol}^{-1}$

F $210\ \text{kJ mol}^{-1}$

WORKED SOLUTION

The products are $60\ \text{kJ mol}^{-1}$ below the reactants. The transition state is $90\ \text{kJ mol}^{-1}$ above the reactants.

Therefore, measured from the products, the transition state is

$$90 + 60 = 150\ \text{kJ mol}^{-1}$$

higher. Hence the reverse activation energy is $\boxed{150\ \text{kJ mol}^{-1}}$, answer **E**.

C-15 Bond energies Answer **C**

QUESTION 15

Hydrogen reacts with chlorine:

$$H_2 + Cl_2 \rightarrow 2HCl.$$

Mean bond energies are:

bond	bond energy / kJ mol^{-1}
H–H	436
Cl–Cl	242
H–Cl	431

What is the approximate enthalpy change for the reaction?

A -678 kJ mol^{-1}

B -242 kJ mol^{-1}

C -184 kJ mol^{-1}

D $+184 \text{ kJ mol}^{-1}$

E $+678 \text{ kJ mol}^{-1}$

WORKED SOLUTION

Energy required to break the bonds in the reactants:

$$436 + 242 = 678 \text{ kJ mol}^{-1}.$$

Energy released when two H–Cl bonds form:

$$2(431) = 862 \text{ kJ mol}^{-1}.$$

Therefore

$$\Delta H = 678 - 862 = -184 \text{ kJ mol}^{-1}.$$

The answer is **C**.

C-16

Calorimetry

Answer B

QUESTION 16

Burning 0.050 mol of a fuel heats 100 g of water by $15°C$.

The specific heat capacity of water is

$$4.2 \text{ J g}^{-1} °\text{C}^{-1}.$$

Only 70% of the energy released by the fuel is transferred to the water.

What is the enthalpy change of combustion of the fuel?

A -252 kJ mol^{-1}

B -180 kJ mol^{-1}

C -126 kJ mol^{-1}

D $+126 \text{ kJ mol}^{-1}$

E $+180 \text{ kJ mol}^{-1}$

F $+252 \text{ kJ mol}^{-1}$

WORKED SOLUTION

The energy absorbed by the water is

$$q = mc\Delta T = 100(4.2)(15) = 6300 \text{ J}.$$

This represents 70% of the energy released, so the fuel releases

$$\frac{6300}{0.70} = 9000 \text{ J} = 9.0 \text{ kJ}$$

when 0.050 mol burns.

Per mole, the magnitude is

$$\frac{9.0}{0.050} = 180\ \mathrm{kJ\ mol^{-1}}.$$

Combustion is exothermic, so $\Delta H = -180\ \mathrm{kJ\ mol^{-1}}$. The answer is **B**.

C-17

Electrolysis

Answer D

QUESTION 17

Inert electrodes are used.

Which row correctly gives the products?

- **A** molten $NaCl$: cathode Na; anode O_2
- **B** dilute $NaCl(aq)$: cathode Na; anode Cl_2
- **C** $CuSO_4(aq)$: cathode H_2; anode Cu
- **D** concentrated $CuCl_2(aq)$: cathode Cu; anode Cl_2
- **E** molten $MgCl_2$: cathode H_2; anode Cl_2

WORKED SOLUTION

In concentrated aqueous copper(II) chloride with inert electrodes, Cu^{2+} is reduced at the cathode:

$$Cu^{2+} + 2e^- \rightarrow Cu.$$

At the anode, chloride ions are oxidised:

$$2Cl^- \rightarrow Cl_2 + 2e^-.$$

So the correct row is **D**.

C-18

Electroplating and concentration

Answer C

QUESTION 18

A $0.250\ dm^3$ solution initially contains 0.100 mol of Cu^{2+}.

Copper is electroplated from the solution until 3.20 g of copper has been deposited. The volume is then restored to $0.250\ dm^3$ by adding water.

$$A_r(Cu) = 64$$

What is the final concentration of Cu^{2+}?

- **A** $0.10\ mol\ dm^{-3}$
- **B** $0.15\ mol\ dm^{-3}$
- **C** $0.20\ mol\ dm^{-3}$
- **D** $0.25\ mol\ dm^{-3}$
- **E** $0.40\ mol\ dm^{-3}$

WORKED SOLUTION

The amount of copper deposited is

$$n(Cu) = \frac{3.20}{64} = 0.0500\ mol.$$

Each mole of deposited copper removes one mole of Cu^{2+}, so the amount remaining is

$$0.100 - 0.0500 = 0.0500\ mol.$$

After the volume is restored to $0.250\ dm^3$,

$$[Cu^{2+}] = \frac{0.0500}{0.250} = 0.20\ mol\ dm^{-3}.$$

The answer is **C**.

C-19

Halogen displacement

Answer A

QUESTION 19

The following reactions occur:

$$X_2 + 2Y^- \rightarrow 2X^- + Y_2$$

and

$$Z_2 + 2X^- \rightarrow 2Z^- + X_2.$$

What is the order of reactivity of the halogens, from most to least reactive?

A $Z > X > Y$

B $X > Z > Y$

C $X > Y > Z$

D $Y > X > Z$

E $Z > Y > X$

WORKED SOLUTION

Because X_2 displaces Y^-, X is more reactive than Y:

$$X > Y.$$

Because Z_2 displaces X^-, Z is more reactive than X:

$$Z > X.$$

Therefore $\boxed{Z > X > Y}$, answer **A**.

C-20

Metal reactivity

Answer C

QUESTION 20

Four metals M, Q, R, T give these observations.

1. Q displaces R from aqueous RSO_4.
2. T also displaces R from aqueous RSO_4.
3. T does not displace Q from aqueous QSO_4.
4. R reacts with dilute hydrochloric acid.
5. M does not react with dilute hydrochloric acid.

Which is the correct order from most reactive to least reactive?

A $Q > R > T > M$

B $T > Q > R > M$

C $Q > T > R > M$

D $Q > T > M > R$

E $R > Q > T > M$

F $T > R > Q > M$

G $M > R > T > Q$

H $Q > M > T > R$

WORKED SOLUTION

Q displaces R, so $Q > R$. T displaces R, so $T > R$. T does not displace Q, so $Q > T$.

Thus $Q > T > R$.

R reacts with hydrochloric acid, so R is above hydrogen in the reactivity series. M does not react with hydrochloric acid, so M is below hydrogen and therefore below R.

The full order is $\boxed{Q > T > R > M}$, answer **C**.

C-21

Fractional distillation

Answer B

QUESTION 21

Hexane has a boiling point of $69^{\circ}C$. Octane has a boiling point of $126^{\circ}C$.

An approximately equal mixture of the liquids is separated using a fractionating column.

When the **first drops of distillate** are collected, which observation is most likely?

- **A** flask below $69^{\circ}C$; top near $69^{\circ}C$
- **B** flask between $69^{\circ}C$ and $126^{\circ}C$; top near $69^{\circ}C$
- **C** flask near $126^{\circ}C$; top near $69^{\circ}C$
- **D** flask between $69^{\circ}C$ and $126^{\circ}C$; top near $126^{\circ}C$
- **E** flask near $126^{\circ}C$; top near $126^{\circ}C$

WORKED SOLUTION

The liquid mixture in the flask boils over a temperature range lying between the boiling points of the two pure components.

Hexane is more volatile, so the vapour reaching the top of an efficient fractionating column is strongly enriched in hexane. When the first distillate is collected, the temperature near the top should therefore be close to $69^{\circ}C$.

Thus **B** is the most likely observation.

C-22

Combustion of hydrocarbons

Answer **C**

QUESTION 22

$25\ cm^3$ of a gaseous alkene is completely burned in excess oxygen.

$100\ cm^3$ of carbon dioxide is produced.

All gas volumes are measured at the same temperature and pressure.

Which is the molecular formula of the alkene?

A C_2H_4

B C_3H_6

C C_4H_8

D C_4H_{10}

E C_5H_{10}

WORKED SOLUTION

At the same temperature and pressure, gas volume ratios equal mole ratios.

The carbon dioxide volume is four times the alkene volume:

$$\frac{100}{25} = 4.$$

One molecule of C_nH_{2n} produces n molecules of CO_2, so $n = 4$.

The alkene is $\boxed{C_4H_8}$, answer **C**.

C-23

Addition polymers

Answer **D**

QUESTION 23

But-1-ene has condensed structural formula

$$CH_2 = CHCH_2CH_3.$$

Which repeating unit represents poly(but-1-ene)?

A $[-CH_2 - CH_2-]_n$

B $[-CH(CH_3) - CH(CH_3)-]_n$

C $[-CH_2 - CH_2 - CH_2 - CH_2-]_n$

D $[-CH_2 - CH(CH_2CH_3)-]_n$

E $[-CH = CH - CH_2 - CH_3-]_n$

WORKED SOLUTION

In addition polymerisation, the carbon–carbon double bond opens to form the polymer backbone. The ethyl group attached to the second alkene carbon remains as a side group.

The repeating unit is therefore

$$\boxed{[-CH_2 - CH(CH_2CH_3)-]_n}.$$

The answer is **D**.

C-24

Esters

Answer **C**

QUESTION 24

A carboxylic acid X reacts with ethanol to form an ester Y and water.

The relative molecular mass of Y is 102.

$$\mathrm{M_r(ethanol)} = 46, \qquad \mathrm{M_r(H_2O)} = 18.$$

What is the relative molecular mass of X?

- **A** 46
- **B** 60
- **C** 74
- **D** 88
- **E** 102
- **F** 120

WORKED SOLUTION

Esterification is

$$\text{acid} + \text{alcohol} \rightarrow \text{ester} + \mathrm{H_2O}.$$

Mass is conserved, so

$$\mathrm{M_r(X)} + 46 = 102 + 18.$$

Hence

$$\mathrm{M_r(X)} = 120 - 46 = 74.$$

The answer is **C**.

C-25

Chemical tests

Answer A

QUESTION 25

An aqueous solution of a salt gives:

- a **green flame**;
- a **blue precipitate** when aqueous sodium hydroxide is added;
- a **white precipitate** when acidified aqueous silver nitrate is added.

Which salt is present?

A $CuCl_2$

B $CuSO_4$

C $FeCl_2$

D $NaCl$

E $CaCl_2$

F $Cu(NO_3)_2$

WORKED SOLUTION

The green/blue-green flame and blue precipitate with sodium hydroxide indicate Cu^{2+}.

The white precipitate with acidified silver nitrate indicates chloride ions, because $AgCl$ is white.

The salt is therefore $\boxed{CuCl_2}$, answer **A**.

C-26

Atmospheric chemistry

Answer E

QUESTION 26

Consider the statements.

1. Carbon monoxide can be formed by incomplete combustion of carbon-containing fuels.
2. Sulfur dioxide can be produced when sulfur-containing impurities in fuels are burned and can contribute to acid rain.
3. Nitrogen oxides are produced mainly because carbon burns with insufficient oxygen.

Which statements are correct?

- **A** none
- **B** 1 only
- **C** 2 only
- **D** 3 only
- **E** 1 and 2 only
- **F** 1 and 3 only
- **G** 2 and 3 only
- **H** 1, 2 and 3

WORKED SOLUTION

Statement 1 is correct: incomplete combustion can produce carbon monoxide.

Statement 2 is correct: sulfur-containing impurities can form sulfur dioxide, which contributes to acid rain.

Statement 3 is false. Nitrogen oxides are formed when nitrogen and oxygen react at high temperatures, for example in combustion engines; they are not mainly a consequence of carbon burning with insufficient oxygen.

Therefore the answer is **E: 1 and 2 only**.

C-27

Quantitative chemistry

Answer **C**

QUESTION 27

A $6.00\ \mathrm{g}$ mixture contains only magnesium and magnesium oxide.

The mixture reacts completely with excess dilute hydrochloric acid.

Exactly $2.40\ \mathrm{dm}^3$ of hydrogen gas is produced at room temperature and pressure.

At rtp,

$$1\ \mathrm{mol\ gas} = 24.0\ \mathrm{dm}^3.$$

$$A_r(\mathrm{Mg}) = 24$$

What percentage by mass of the original mixture was magnesium?

- **A** 20%
- **B** 30%
- **C** 40%
- **D** 50%
- **E** 60%

WORKED SOLUTION

Only magnesium metal produces hydrogen with dilute hydrochloric acid. Magnesium oxide neutralises the acid but does not form hydrogen.

The amount of hydrogen formed is

$$n(H_2) = \frac{2.40}{24.0} = 0.100 \text{ mol}.$$

From

$$Mg + 2HCl \rightarrow MgCl_2 + H_2,$$

the amount of magnesium is also 0.100 mol. Its mass is

$$0.100(24) = 2.40 \text{ g}.$$

Therefore

$$\%Mg = \frac{2.40}{6.00} \times 100 = 40\%.$$

The answer is **C**.

LEVEL 1 · STANDARD

Biology

27 QUESTIONS · COMPLETE SOLUTIONS

Preserved Biology Standard source set. Question wording, option order, answer and solution method are unchanged.

B-01

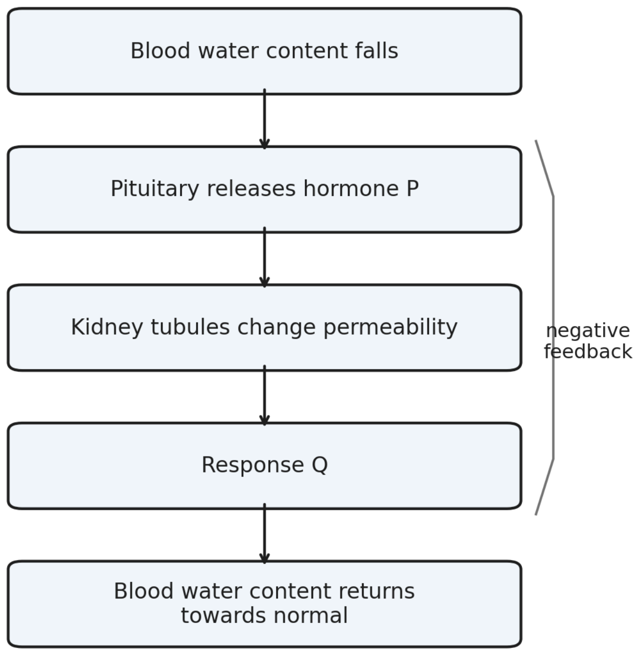
Homeostasis

Answer D

QUESTION 1

A person loses water by sweating for a prolonged period.

The diagram summarises part of the homeostatic response.



A insulin; more water reabsorbed by the kidneys

B glucagon; less water reabsorbed by the kidneys

C ADH; less water reabsorbed and a larger volume of urine produced

D ADH; more water reabsorbed and a smaller volume of urine produced

E adrenaline; less water reabsorbed and more concentrated urine

WORKED SOLUTION

Low blood water content causes increased release of ADH from the pituitary gland.

ADH increases the permeability of parts of the kidney tubule, so more water is reabsorbed into the blood. Less water is therefore lost in urine, giving a smaller volume of more concentrated urine.

Hence the correct answer is **D**.

B-02

Cell structure

Answer **C**

QUESTION 2

A cell has:

- a cell membrane;
- cytoplasm;
- a cell wall;
- circular chromosomal DNA;
- several small plasmids;
- no mitochondria.

Which statement about the cell is correct?

A It must contain chloroplasts.

B It cannot carry genetic information.

C A plasmid may carry a gene for antibiotic resistance.

D It is a eukaryotic cell.

E It cannot carry out respiration.

WORKED SOLUTION

The combination of circular chromosomal DNA, plasmids and absence of mitochondria is characteristic of a bacterial cell.

Plasmids can carry additional genes, including genes that confer antibiotic resistance. Bacteria can still carry out respiration even though they do not possess mitochondria.

Therefore the correct answer is **C**.

B-03 DNA and base composition Answer C

QUESTION 3

A double-stranded DNA molecule is **300 base pairs** long.

Adenine accounts for 36% of all bases in the molecule.

How many guanine bases are present?

A 42

B 72

C 84

D 108

E 168

WORKED SOLUTION

There are

$$300 \times 2 = 600$$

bases in total.

In double-stranded DNA, adenine and thymine occur in equal proportions. Therefore

$$A + T = 36\% + 36\% = 72\%.$$

The remaining 28% consists equally of cytosine and guanine, so guanine makes up 14%.

$$0.14 \times 600 = 84.$$

Thus the correct answer is **C**.

B-04

Natural selection

Answer G

QUESTION 4

A population of beetles contains both pale and dark individuals.

Before an industrial development, most tree bark in the beetles' habitat is pale. After many years of soot deposition, most bark becomes much darker. The frequency of an allele associated with dark colour subsequently increases.

Which statements could explain the change?

1. Soot caused pale beetles to mutate specifically into dark beetles.
2. Dark beetles may have become less visible to predators.
3. If colour is heritable, dark beetles surviving and reproducing more successfully could increase the frequency of the dark allele.

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Mutation is not directed by environmental need, so statement 1 is incorrect.

Dark beetles could be better camouflaged on darkened bark, increasing their chance of survival. If the difference is heritable, greater reproductive success of dark beetles can increase the frequency of the associated allele.

Statements 2 and 3 are therefore correct, giving **G**.

B-05

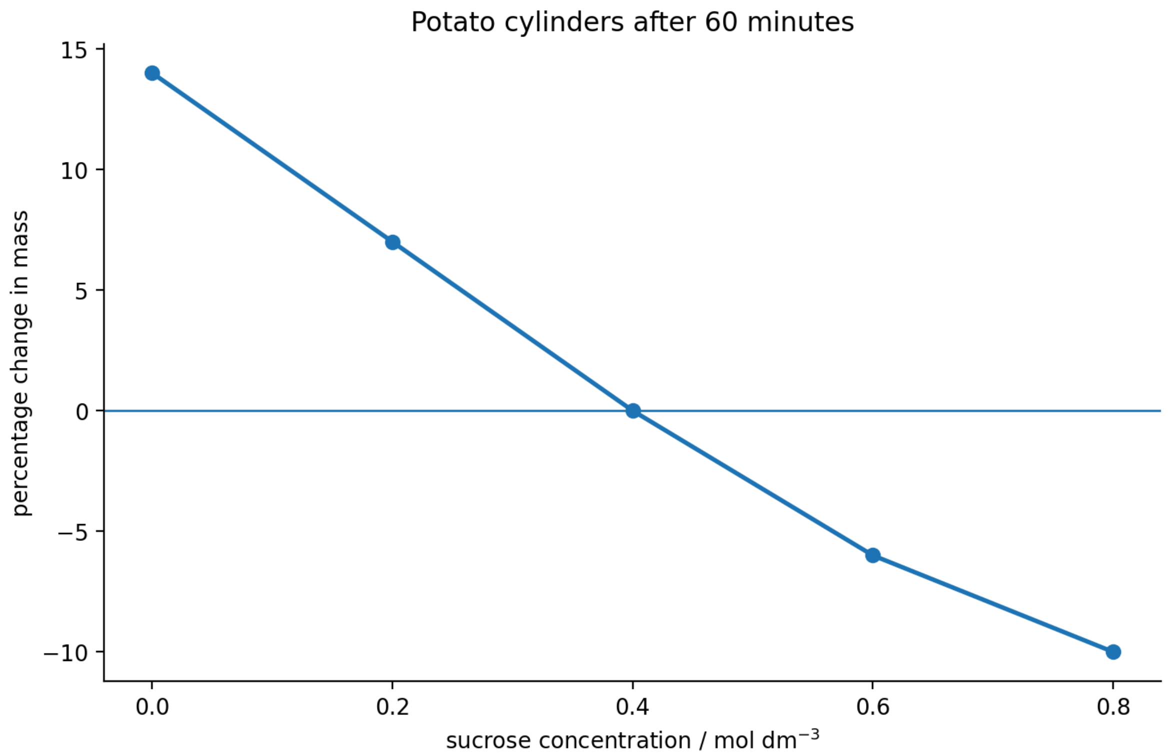
Osmosis

Answer **C**

QUESTION 5

Identical potato cylinders were placed for 60 minutes in different sucrose solutions.

Which row is correct?



	concentration giving approximately no net change in mass / mol dm^{-3}	net movement of water at 0.20 mol dm^{-3}
A	0.20	into the potato cells
B	0.40	out of the potato cells
C	0.40	into the potato cells
D	0.60	into the potato cells
E	0.40	no net movement

A row A

B row B

C row C

D row D

E row E

WORKED SOLUTION

At approximately $0.40\ \mathrm{mol\ dm^{-3}}$, the potato cylinders show no percentage change in mass, indicating no net movement of water.

At $0.20\ \mathrm{mol\ dm^{-3}}$, the cylinders gain mass, so there has been a net movement of water into the potato cells by osmosis.

Therefore row **C** is correct.

B-06

Enzymes

Answer **F**

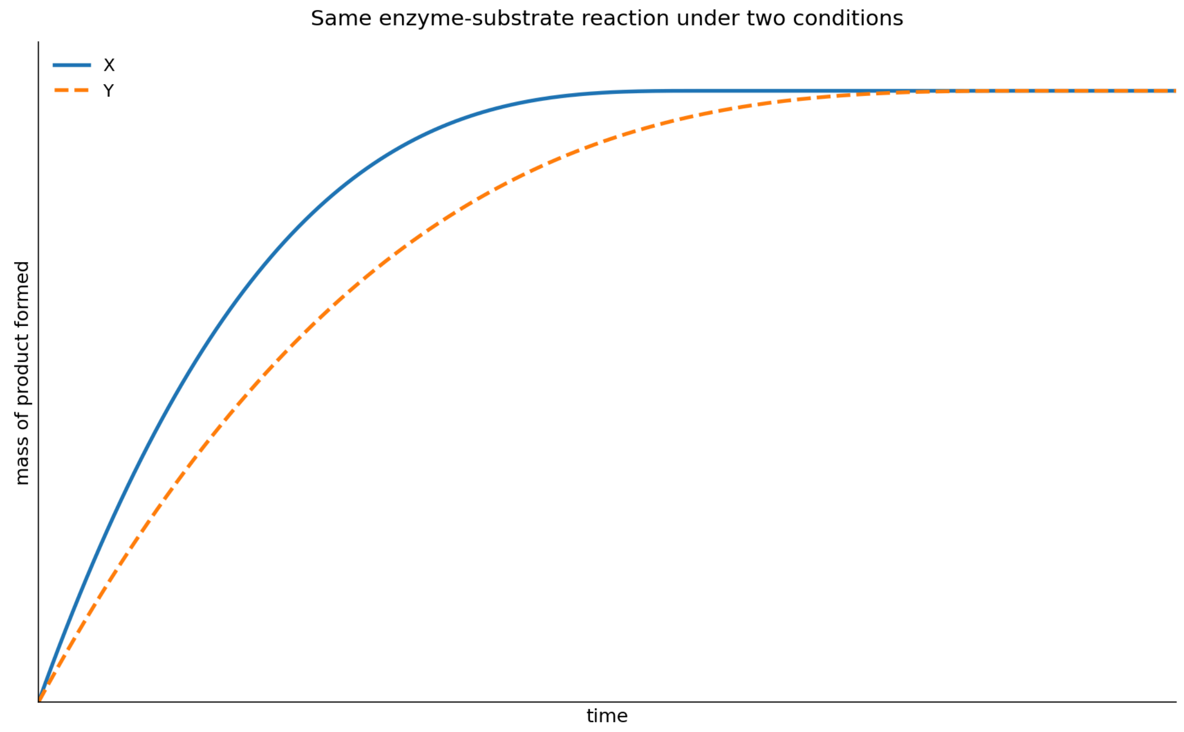
QUESTION 6

The graph shows the mass of product produced during the same enzyme-controlled reaction under two different conditions, X and Y.

The graph shows both reactions until no further product is formed.

Which changes **could** explain the difference between X and Y?

1. X contains a greater concentration of enzyme than Y.
2. X contains a greater initial amount of substrate than Y.
3. The temperature in X is closer to the enzyme's optimum than the temperature in Y, with neither temperature causing denaturation.



A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Both curves reach the same final mass of product, so the graph does not support X having a greater initial amount of substrate than Y.

A greater enzyme concentration can increase the rate of reaction without changing the total amount of substrate ultimately converted. A temperature closer to the enzyme's optimum can also increase the rate, provided neither temperature causes denaturation.

Therefore statements 1 and 3 could explain the graph: **F**.

B-07

Cell division

Answer E

QUESTION 7

A human body cell contains 46 chromosomes and $6\ \mathrm{pg}$ of DNA before DNA replication.

Which row correctly describes the same cell immediately before mitosis and one daughter cell after mitosis?

	immediately before mitosis: chromosomes	immediately before mitosis: DNA / pg	one daughter cell after mitosis: chromosomes	one daughter cell after mitosis: DNA / pg
A	23	12	23	6
B	46	6	46	6
C	92	12	46	6
D	46	12	23	6
E	46	12	46	6

A row A

B row B

C row C

D row D

E row E

WORKED SOLUTION

DNA replication doubles the amount of DNA from $6\ \mathrm{pg}$ to $12\ \mathrm{pg}$, but the chromosome number is still counted as 46 because each chromosome consists of two sister chromatids.

After mitosis, each daughter cell has the normal diploid chromosome number and half the replicated DNA:

$$46\ \text{chromosomes}, \qquad 6\ \text{pg DNA}.$$

So row **E** is correct.

B-08

Stem cells

Answer F

QUESTION 8

Which statements about an adult stem cell from human bone marrow are correct?

1. It may divide by mitosis to produce more stem cells.
2. It is totipotent and can form an entire new human organism.
3. It can differentiate into a limited range of specialised blood cells.

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Adult stem cells can divide by mitosis and self-renew. Bone-marrow stem cells are multipotent: they can form a restricted range of specialised blood cells.

They are not totipotent and cannot normally form an entire new organism.

Therefore statements 1 and 3 only are correct: **F**.

B-09

Inheritance and pedigrees

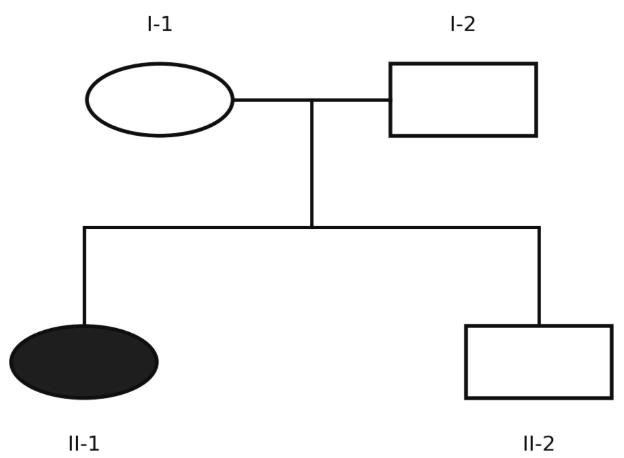
Answer C

QUESTION 9

The condition shown in the pedigree is caused by an **autosomal recessive allele**.

Individual II-2 is unaffected.

If II-2 later has a child with a woman who is affected by the condition, what is the probability that their child will be affected?



Filled symbol = affected by an autosomal recessive condition

A $\frac{1}{6}$

B $\frac{1}{4}$

C $\frac{1}{3}$

D $\frac{1}{2}$

E $\frac{2}{3}$

WORKED SOLUTION

Because two unaffected parents produced an affected child with a recessive condition, both parents must be heterozygous:

$$\mathrm{Aa} \times \mathrm{Aa}.$$

Given that II-2 is unaffected, the possible genotypes are $\mathrm{AA}, \mathrm{Aa}, \mathrm{Aa}$. Therefore

$$\mathrm{P}\,(\text{II-2 is Aa}) = \frac{2}{3}.$$

The affected woman is aa. If II-2 is a carrier, the probability that their child is affected is $\frac{1}{2}$.

Hence

$$\frac{2}{3} \times \frac{1}{2} = \boxed{\frac{1}{3}}.$$

The correct answer is **C**.

B-10

Monohybrid inheritance

Answer **D**

QUESTION 10

In rabbits, black fur is caused by a dominant allele B, while brown fur is caused by the recessive allele b.

Two black rabbits produce a brown offspring.

What is the probability that their **next two offspring are both black**?

A $\frac{1}{4}$

B $\frac{3}{8}$

C $\frac{1}{2}$

D $\frac{9}{16}$

E $\frac{3}{4}$

WORKED SOLUTION

A brown offspring must have genotype bb, so each black parent must have contributed a recessive allele. The parental cross is therefore

$$Bb \times Bb.$$

The probability that one offspring is black is $\frac{3}{4}$.

For two successive offspring:

$$\frac{3}{4} \times \frac{3}{4} = \boxed{\frac{9}{16}}.$$

Therefore the answer is **D**.

B-11 DNA and mutation Answer **B**

QUESTION 11

The relevant coding-DNA triplets are shown.

A section of coding DNA is

$$\text{ATG CCT GAA TTT}.$$

A substitution changes the second triplet from CCT to CAT.

Which statements are correct?

1. Exactly one amino acid in the section is changed.
2. The reading frame is shifted.
3. The number of amino acids coded by the shown section must decrease.

coding-DNA triplet	amino acid
ATG	methionine
CCT	proline
CAT	histidine
GAA	glutamate
TTT	phenylalanine

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

The substitution changes the triplet from CCT to CAT, which changes proline to histidine.

A substitution neither inserts nor removes a base, so the reading frame is not shifted. No information given indicates that a stop signal has been introduced.

Thus only statement 1 is correct: **B**.

B-12

Gene technology

Answer E

QUESTION 12

A suitable copy of a human gene is inserted into a bacterial plasmid so that bacteria can produce the corresponding human protein.

Which statements are correct?

1. A restriction enzyme may be used to cut DNA.
2. DNA ligase may be used to join the human gene to plasmid DNA.
3. The recombinant plasmid must enter the nucleus of the bacterial cell before the gene can function.

- **A** none
- **B** 1 only
- **C** 2 only
- **D** 3 only
- **E** 1 and 2 only
- **F** 1 and 3 only
- **G** 2 and 3 only
- **H** 1, 2 and 3

WORKED SOLUTION

Restriction enzymes can cut DNA at particular sequences, and DNA ligase can join the suitable human gene copy to plasmid DNA.

Bacteria do not possess a nucleus, so the recombinant plasmid does not have to enter one.

Therefore statements 1 and 2 only are correct: **E**.

B-13

Respiration during exercise

Answer **B**

QUESTION 13

The table shows oxygen demand and oxygen supplied to working muscle at different times during exercise.

Here, oxygen demand means the amount of oxygen that would be required to meet the muscle's energy demand entirely by aerobic respiration.

Assume the oxygen supplied is all available for aerobic respiration.

At which measured times must anaerobic respiration also be contributing to the muscle's energy supply?

time / min	oxygen demand / arbitrary units	oxygen supplied / arbitrary units
0	0.8	0.8
2	1.4	1.0
4	1.8	1.3
6	2.0	1.7
8	1.6	1.6
10	1.1	1.2

A 0 only

B 2, 4 and 6 only

C 8 only

D 2, 4, 6 and 8

E 10 only

WORKED SOLUTION

When this oxygen demand exceeds the oxygen supplied, aerobic respiration alone cannot meet the muscle's stated energy demand, so anaerobic respiration must also contribute.

This occurs at 2, 4 and 6 minutes.

At 8 minutes demand equals supply, and at 10 minutes supply is greater than demand.

Therefore the answer is **B**.

B-14 Circulation Answer **D**

QUESTION 14

Three blood vessels have the following properties.

- Vessel P carries blood **from the heart to the lungs** and has relatively low oxygen concentration.
- Vessel Q carries blood **from the lungs to the heart** and has relatively high oxygen concentration.
- Vessel R carries oxygenated blood **from the heart to a kidney**.

Which row correctly identifies the vessels?

	P	Q	R
A	pulmonary vein	pulmonary artery	renal vein
B	aorta	vena cava	pulmonary artery
C	pulmonary artery	pulmonary vein	renal vein
D	pulmonary artery	pulmonary vein	renal artery
E	vena cava	pulmonary artery	aorta

A row A

B row B

C row C

D row D

E row E

WORKED SOLUTION

The pulmonary artery carries relatively deoxygenated blood from the heart to the lungs.

The pulmonary vein carries oxygenated blood from the lungs to the heart.

The renal artery carries oxygenated blood from the heart to a kidney.

Therefore row **D** is correct.

B-15 Digestion Answer **C**

QUESTION 15

Four tubes are incubated at $37^{\circ}\mathrm{C}$.

The pH of each tube is monitored continuously.

The lipase used has an optimum near pH 8. Bile emulsifies lipid into smaller droplets but contains no digestive enzyme.

Which tube would be expected to show the **greatest initial rate of decrease in pH**?

tube	contents
A	lipid + water
B	lipid + lipase at pH 8
C	lipid + lipase + bile at pH 8
D	lipid + lipase + bile at pH 2

A A

B B

C C

D D

E All at the same rate

WORKED SOLUTION

Lipase hydrolyses lipids to products including fatty acids, so lipase activity causes the pH to fall.

Tubes B and C are at pH 8, near the stated optimum for the lipase. Tube C also contains bile, which emulsifies the lipid into smaller droplets and increases the surface area available to the enzyme. Tube D is at pH 2, far from the stated optimum, and tube A contains no lipase.

Tube C therefore has the conditions expected to give the greatest initial rate of decrease in pH. The answer is **C**.

B-16

Kidney function

Answer E

QUESTION 16

The table gives relative concentrations of substances in three fluids from a healthy person.

Which statements are correct?

1. Most plasma proteins are prevented from entering the filtrate.
2. Glucose is reabsorbed from the nephron.
3. The greater concentration of urea in urine proves that urea is produced by kidney cells.

substance	blood plasma	filtrate entering nephron	urine
protein	8.0	0	0
glucose	1.0	1.0	0
urea	0.3	0.3	2.0

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

The absence of protein from the filtrate is consistent with large plasma proteins being retained in the blood.

Glucose is present in the filtrate but absent from normal urine, supporting its reabsorption.

A greater urea concentration in urine does not show that the kidney produces urea; water removal can concentrate urea already present.

Therefore statements 1 and 2 only are correct: **E**.

B-17

Nervous system

Answer **E**

QUESTION 17

Which statements about a simple reflex arc are correct?

1. A sensory neurone can carry an impulse from a receptor towards the central nervous system.
2. A relay neurone can connect a sensory neurone to a motor neurone within the central nervous system.
3. A motor neurone normally carries impulses from an effector towards the central nervous system.

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Sensory neurones carry impulses from receptors towards the CNS. Relay neurones can connect sensory and motor neurones within the CNS.

Motor neurones carry impulses from the CNS to effectors, not from effectors back towards the CNS.

Therefore statements 1 and 2 only are correct: **E**.

B-18 Disease and body defence Answer **H**

QUESTION 18

Which statements are correct?

1. Antibiotics do not directly destroy viruses.
2. Vaccination may lead to the formation of memory cells.
3. Antibiotic treatment can favour the survival and reproduction of resistant bacteria already present in a population.

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

All three statements are correct.

Antibiotics target bacteria rather than directly destroying viruses. Vaccination can produce memory cells. Antibiotic exposure also acts as a selection pressure, allowing resistant bacteria to survive and reproduce more successfully.

Therefore the answer is **H**.

B-19 Ecosystems Answer B

QUESTION 19

Students record four species along a transect.

Which statements are correct?

1. Species richness is greatest at both 10 m and 20 m.
2. Total abundance is greatest at 30 m.
3. Species C occurs at every sampled position.

distance / m	species A	species B	species C	species D
0	4	0	2	0
10	3	1	2	0
20	0	4	3	1
30	0	2	0	5

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Species richness is the number of different species present. At both 10 m and 20 m, three species are present, the greatest richness recorded.

Total abundance is $6, 6, 8, 7$ at 0, 10, 20 and 30 m respectively, so statement 2 is false.

Species C is absent at 30 m, so statement 3 is false.

Thus statement 1 only is correct: **B**.

B-20 Reproductive hormones Answer **C**

QUESTION 20

Hormones P, Q and R have the following roles.

- P is released from the pituitary gland and stimulates development of an ovarian follicle.
- Q is released from the pituitary gland and its rise triggers ovulation.
- R is released from the ovary after ovulation and helps maintain the uterine lining.

Which row correctly identifies P, Q and R?

	P	Q	R
A	FSH	progesterone	LH
B	LH	FSH	progesterone
C	FSH	LH	progesterone
D	oestrogen	LH	FSH
E	FSH	LH	oestrogen

A row A

B row B

C row C

D row D

E row E

WORKED SOLUTION

FSH stimulates development of an ovarian follicle. LH triggers ovulation. Progesterone is produced after ovulation and helps maintain the uterine lining.

Therefore the correct row is **C**.

B-21

Photosynthesis and limiting factors

Answer **E**

QUESTION 21

A plant is exposed to different light intensities and carbon dioxide concentrations. All other conditions are kept constant.

Which statements are supported by the data?

1. At low light intensity, light could be the limiting factor.
2. At high light intensity and 0.04% carbon dioxide, carbon dioxide could be a limiting factor.
3. Increasing light intensity and carbon dioxide concentration indefinitely must continue increasing the rate.

light intensity	carbon dioxide concentration	rate of photosynthesis / arbitrary units
low	0.04%	5
low	0.10%	5
high	0.04%	8
high	0.10%	14

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

At low light intensity, increasing carbon dioxide does not increase the rate, so light could be limiting.

At high light intensity, increasing carbon dioxide from 0.04% to 0.10% increases the rate, so carbon dioxide could be limiting at 0.04%.

The rate cannot be assumed to increase indefinitely because another factor will eventually become limiting.

Therefore statements 1 and 2 only are correct: **E**.

B-22

Transpiration

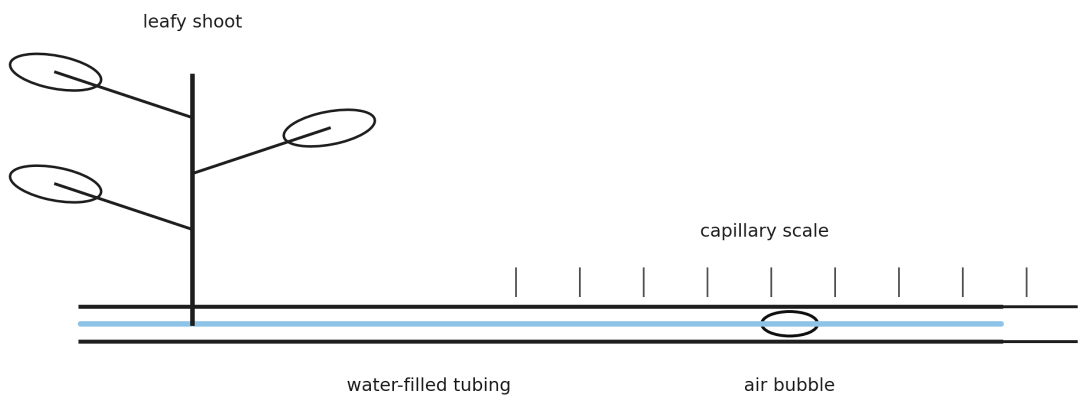
Answer **D**

QUESTION 22

A potometer is used to estimate water uptake by a leafy shoot.

The air bubble moves 18 mm in 6 minutes. The cross-sectional area of the capillary tube is 0.50 mm^2.

Which row gives the rate of water uptake and the likely effect of increasing the humidity around the leaves?



	water uptake / $mm^3\ min^{-1}$	movement of bubble when humidity increases
A	0.75	faster
B	0.75	slower
C	1.50	faster
D	1.50	slower
E	3.00	slower

A row A

B row B

C row C

D row D

E row E

WORKED SOLUTION

The volume of water taken up is

$$18 \times 0.50 = 9.0\ \mathrm{mm}^3.$$

Over 6 minutes, the rate is

$$\frac{9.0}{6} = 1.50\ \mathrm{mm}^3\ \mathrm{min}^{-1}.$$

Increasing humidity reduces the water-vapour concentration gradient between the leaf and surrounding air, so transpiration decreases and the bubble moves more slowly.

Therefore row **D** is correct.

B-23

Translocation

Answer **B**

QUESTION 23

A plant is exposed to carbon dioxide containing radioactive carbon.

Several hours later, radioactive sucrose is detected in the roots.

Which explanation is correct?

- **A** Radioactive carbon dioxide entered the roots and was converted directly to sucrose.
- **B** Radioactive carbon dioxide entered photosynthetic tissue, was incorporated into sugars, and some sugar was transported to the roots through the phloem.
- **C** Radioactive carbon dioxide entered the leaves and was transported to the roots through the xylem before photosynthesis occurred.
- **D** Radioactive water formed in the leaves and moved through the phloem.
- **E** Radioactive carbon dioxide was absorbed directly by root hair cells by active transport.

WORKED SOLUTION

Carbon dioxide enters photosynthetic tissue and is incorporated into organic compounds during photosynthesis.

Sugars such as sucrose can then move from source tissues such as leaves to sink tissues such as roots by translocation through the phloem.

Therefore the correct answer is **B**.

B-24

Root uptake

Answer B

QUESTION 24

Root hair cells contain a higher concentration of nitrate ions than the surrounding soil solution.

Nitrate ions nevertheless continue to enter the cells.

When a chemical that inhibits respiration is added, nitrate uptake falls sharply, but water can initially continue to enter the cells.

Which row best explains these observations?

	nitrate-ion uptake	water uptake
A	diffusion	active transport
B	active transport	osmosis
C	diffusion	diffusion
D	osmosis	diffusion
E	active transport	active transport

A row A

B row B

C row C

D row D

E row E

WORKED SOLUTION

Nitrate ions are moving into cells despite already being at a higher concentration inside, so uptake is against the concentration gradient and requires active transport.

Respiration provides energy for this process, explaining why the inhibitor reduces nitrate uptake.

Water enters cells across a partially permeable membrane by osmosis.

Therefore row **B** is correct.

B-25

Surface area to volume ratio

Answer **C**

QUESTION 25

A cube of living tissue has sides of $3\ \mathrm{mm}$.

It is cut into 27 separate cubes, each with sides of $1\ \mathrm{mm}$. The total volume of tissue is unchanged.

Which row correctly describes the effect?

	total surface area after cutting compared with before	maximum diffusion distance to the centre of a cube
A	unchanged	decreases
B	doubles	unchanged
C	triples	decreases
D	increases sixfold	increases
E	increases ninefold	decreases

A row A

B row B

C row C

D row D

E row E

WORKED SOLUTION

The original surface area is

$$6(3^2) = 54\ \mathrm{mm}^2.$$

Each $1\ \mathrm{mm}$ cube has surface area $6\ \mathrm{mm}^2$, so 27 cubes have total surface area

$$27(6) = 162\ \mathrm{mm}^2.$$

Thus surface area triples:

$$\frac{162}{54} = 3.$$

The maximum distance from the surface to the centre falls from $1.5\ \mathrm{mm}$ to $0.5\ \mathrm{mm}$.

Therefore row **C** is correct.

B-26 Meiosis and sex determination | Answer **F**

QUESTION 26

A cell in the testes of a healthy human male undergoes meiosis to produce sperm.

Which statements are correct?

1. Each normal sperm produced contains 23 chromosomes.

2. Every sperm produced contains a Y chromosome.

3. Fertilisation of an egg by a Y-bearing sperm can produce an XY zygote.

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Meiosis produces haploid gametes containing 23 chromosomes.

Sperm contain either an X chromosome or a Y chromosome, so statement 2 is false.

A Y-bearing sperm fertilising an X-bearing egg can produce an XY zygote.

Therefore statements 1 and 3 only are correct: **F**.

B-27

Blood glucose control

Answer **E**

QUESTION 27

After a carbohydrate-rich meal, the blood glucose concentration of a healthy person rises.

Which statements are correct?

1. Insulin secretion should increase.
2. Increased uptake of glucose by target tissues can contribute to the return of blood glucose concentration towards its normal level.
3. Glucagon secretion should increase because glucagon promotes uptake of glucose from the blood into the liver.

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

A rise in blood glucose concentration stimulates increased insulin secretion.

Insulin promotes processes that remove glucose from the blood, helping return the concentration towards its normal range by negative feedback.

Glucagon acts in the opposite direction when blood glucose is low; it promotes processes that increase blood glucose concentration.

Therefore statements 1 and 2 only are correct: **E**.

Compilation record: Level 1 uses Engineering Mock Test 3 for Mathematics/Physics and the corresponding Standard Biology/Chemistry source sets. This solution book is ready to support the later construction of the matching full test.